

KEVIN REN

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Princeton PhD working on high-dimensional probability, geometric analysis, and algorithms, with applications to modern machine learning and statistics.

EDUCATION

Princeton University, Princeton, NJ

- Ph.D. in Mathematics (Advisor: Assaf Naor)
- M.S. in Mathematics

Expected May 2027
Sep 2023

Massachusetts Institute of Technology, Cambridge, MA

- B.S. in Mathematics and Physics (GPA 5.0/5.0)

May 2022

RESEARCH & INDUSTRY EXPERIENCE

Graduate Researcher, Princeton University

2022–Present

- Research in geometry, high-dimensional probability, and algorithms with connections to ML theory.
- 17 papers/preprints, including STOC 2025 and SODA 2026; work featured twice in *Quanta Magazine*.
- Developed probabilistic and geometric tools for embeddings, growth bounds, stability, and structure recovery.

Quantitative Research Intern, Jane Street Capital

Summer 2024

- Analyzed the market impact of high-volatility events (like company earnings reports).
- Designed and implemented a rigorous Python pipeline to evaluate predictive signals on financial data.
- Rapidly iterated on signal-finding methods using Bayesian reasoning and hypothesis testing.

Visiting Researcher, Alignment Research Center

Summer 2026

- Learned about ARC's goal of designing principled algorithms to understand the behavior of powerful AI models.
- Showed empirical and theoretical guarantees for the toy model of a random MLP with ReLU activations.
- Discussed ideas with mentors, fellow researchers, and LLMs, and used the feedback to hone my ideas.

SELECTED PROJECT

Manifold Reconstruction from Noisy Data

- Designed a new geometry-aware, statistical algorithm to identify low-dimensional manifold structure from noisy data.
- Presented at 2025 Harvard Conference on Geometry and Statistics.

HONORS & AWARDS

Simons Foundation Dissertation Fellowship for Mathematics

2025–27

Cubist / Point72 PhD Fellowship

2025–26

NSF Graduate Research Fellowship (GRFP)

2022–27

Alibaba Global Math Competition, Global Top 70

2021–23

Putnam Top 25

2021

USA Math Olympiad Winner (Akamai Award)

2015, 2016

SKILLS

Programming: Python (numpy/pandas), PyTorch, C++, MATLAB, Mathematica

Languages: English (Native), Chinese (Fluent)